

EEGLAB Practical Exercise 02

Session: ICA

Name & Family Name

Date

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sep16 2026

PART 1 | Using project03.set raw data set

1. Preprocess the data using the EEGLAB pipeline. (Channel location, Remove bad channel if exists, Filtering (0.5_45), Removing line noise, Resample and re-reference if necessary, Artifact rejection if needed)

#1 Sorting channels

Standard-10-20-Cap64.ced

Standard-10-20-Cap64 (1).ced

+

File

Edit

View

Number	labels	theta	radius	X	Y	Z	sph_theta	sph_phi	sph_radius	
1	FT9	-72	0.639	0.28	0.862	-0.423	72	-25	1	
2	F7	-54	0.511	0.587	0.809	-0.0349	54	-2	1	
3	FT7	-72	0.511	0.309	0.95	-0.0349	72	-2	1	
4	T7	-90	0.511	6.12e-17		0.999	-0.0349	90	-2	1
5	AF7	-38	0.511	0.788	0.615	-0.0349	38	-2	1	
6	F5	-49	0.417	0.634	0.729	0.259	49	15	1	
7	FC5	-69	0.394	0.339	0.883	0.326	69	19	1	
8	C5	-90	0.383	5.72e-17		0.934	0.358	90	21	1
9	FP1	-18	0.511	0.95	0.309	-0.0349	18	-2	1	
10	AF3	-23	0.411	0.885	0.376	0.276	23	16	1	
11	F3	-39	0.333	0.673	0.545	0.5	39	30	1	
12	FC3	-62	0.278	0.36	0.676	0.643	62	40	1	
13	F1	-22	0.278	0.71	0.287	0.643	22	40	1	
14	FC1	-45	0.178	0.375	0.375	0.848	45	58	1	
15	FZ	0	0.256	0.719	-0	0.695	-0	44	1	
16	FCz	0	0.128	0.391	-0	0.921	-0	67	1	
17	TP7	-108	0.511	-0.309	0.95	-0.0349	108	-2	1	
18	P7	-126	0.511	-0.587	0.809	-0.0349	126	-2	1	
19	PO9	-144	0.639	-0.733	0.533	-0.423	144	-25	1	
20	CP5	-111	0.394	-0.339	0.883	0.326	111	19	1	
21	P5	-131	0.417	-0.634	0.729	0.259	131	15	1	
22	PO7	-144	0.511	-0.809	0.587	-0.0349	144	-2	1	
23	C3	-90	0.256	4.4e-17	0.719	0.695	90	44	1	
24	CP3	-118	0.278	-0.36	0.676	0.643	118	40	1	
25	O1	-162	0.511	-0.95	0.309	-0.0349	162	-2	1	
26	Oz	180	0.511	-0.999	-1.22e-16	-0.0349	-180	-2	1	
27	P3	-141	0.333	-0.673	0.545	0.5	141	30	1	
28	PO3	-157	0.411	-0.885	0.376	0.276	157	16	1	
29	POz	180	0.383	-0.934	-1.14e-16	0.358	-180	21	1	
30	P1	-158	0.278	-0.71	0.287	0.643	158	40	1	
31	PZ	180	0.256	-0.719	-8.81e-17	0.695	-180	44	1	
32	CP1	-135	0.178	-0.375	0.375	0.848	135	58	1	
33	FT10	72	0.639	0.28	-0.862	-0.423	-72	-25	1	
34	F8	54	0.511	0.587	-0.809	-0.0349	-54	-2	1	
35	FT8	72	0.511	0.309	-0.95	-0.0349	-72	-2	1	
36	T8	90	0.511	6.12e-17		-0.999	-0.0349	-90	-2	1
37	AF8	38	0.511	0.788	-0.615	-0.0349	-38	-2	1	
38	F6	49	0.417	0.634	-0.729	0.259	-49	15	1	
39	FC6	69	0.394	0.339	-0.883	0.326	-69	19	1	

Ln 11, Col 1

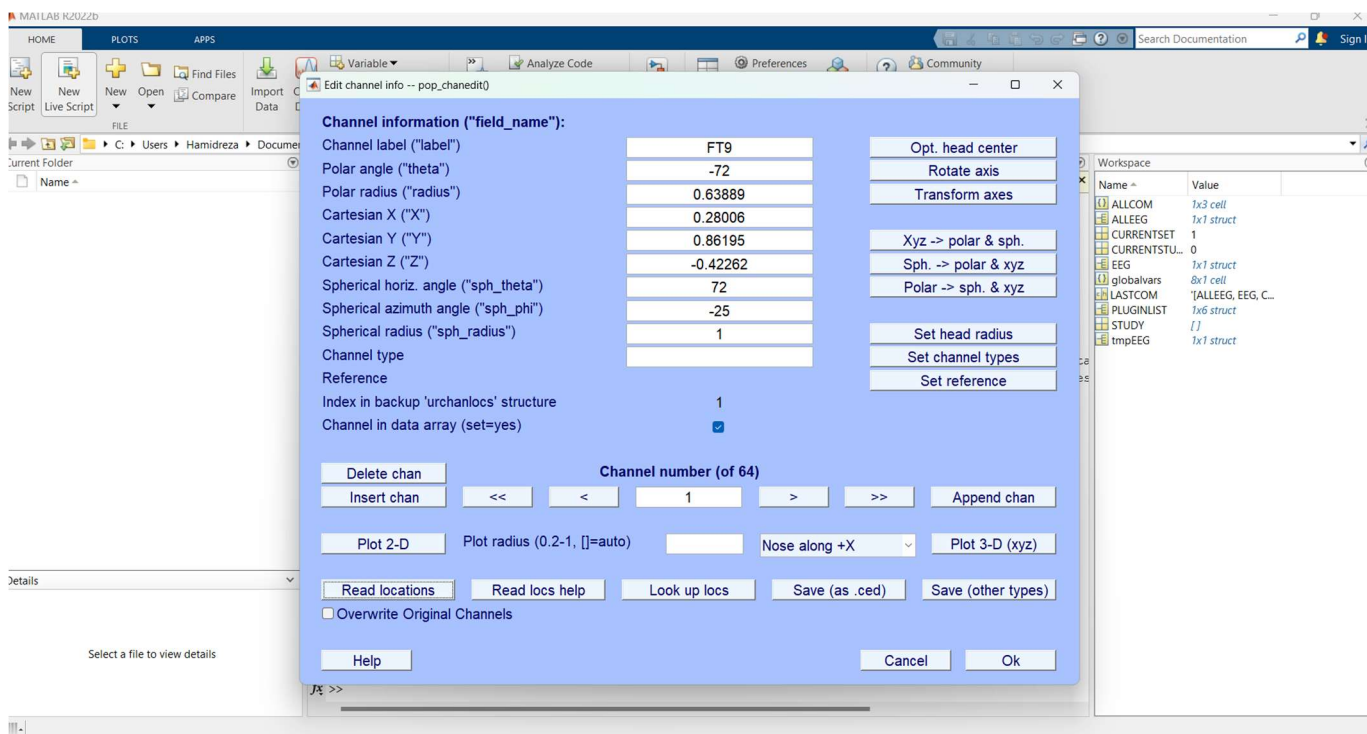
46 of 3234 characters

Plain text

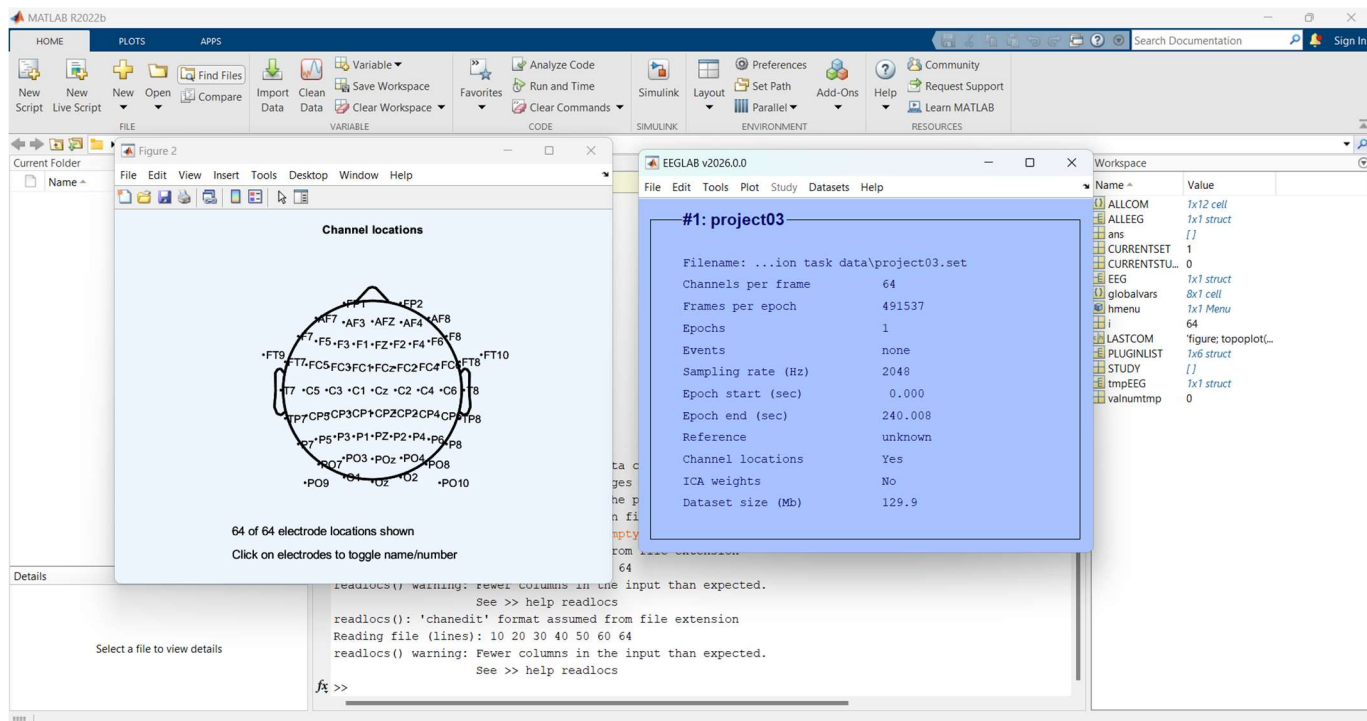
100%

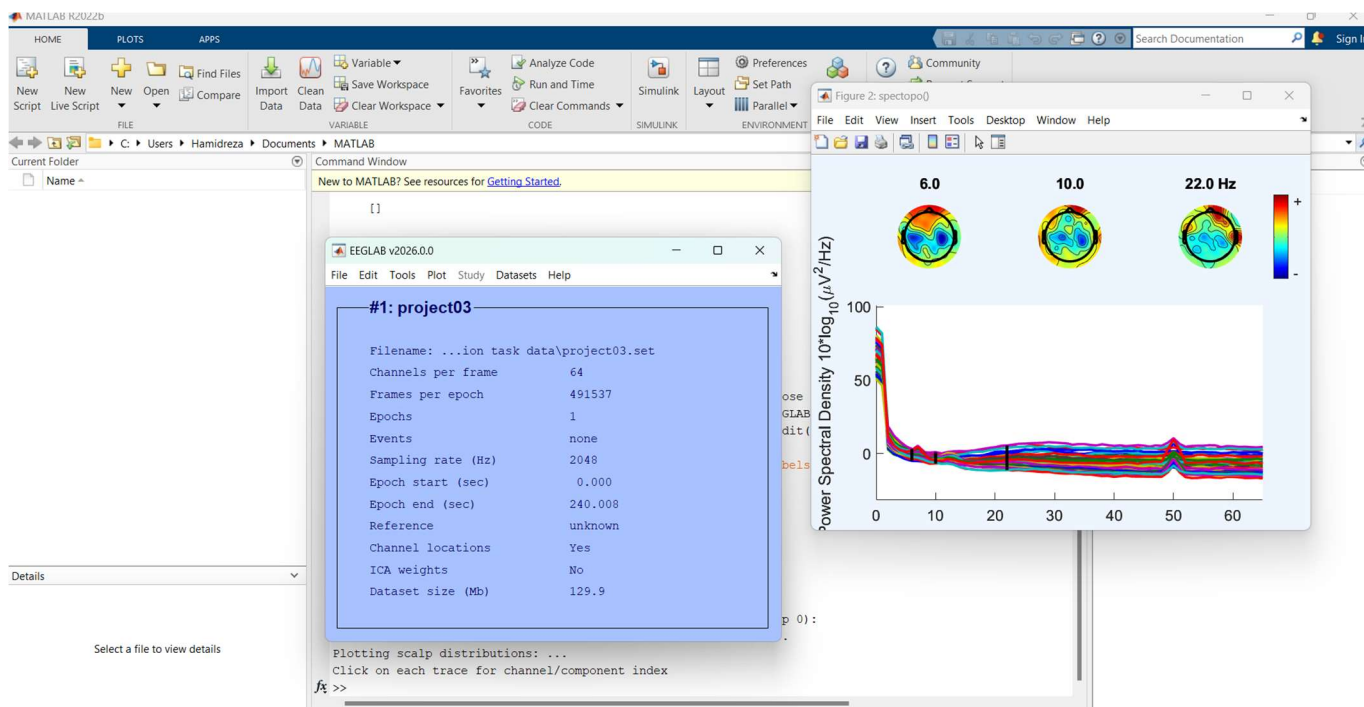
Windows (CRLF)

UTF-8

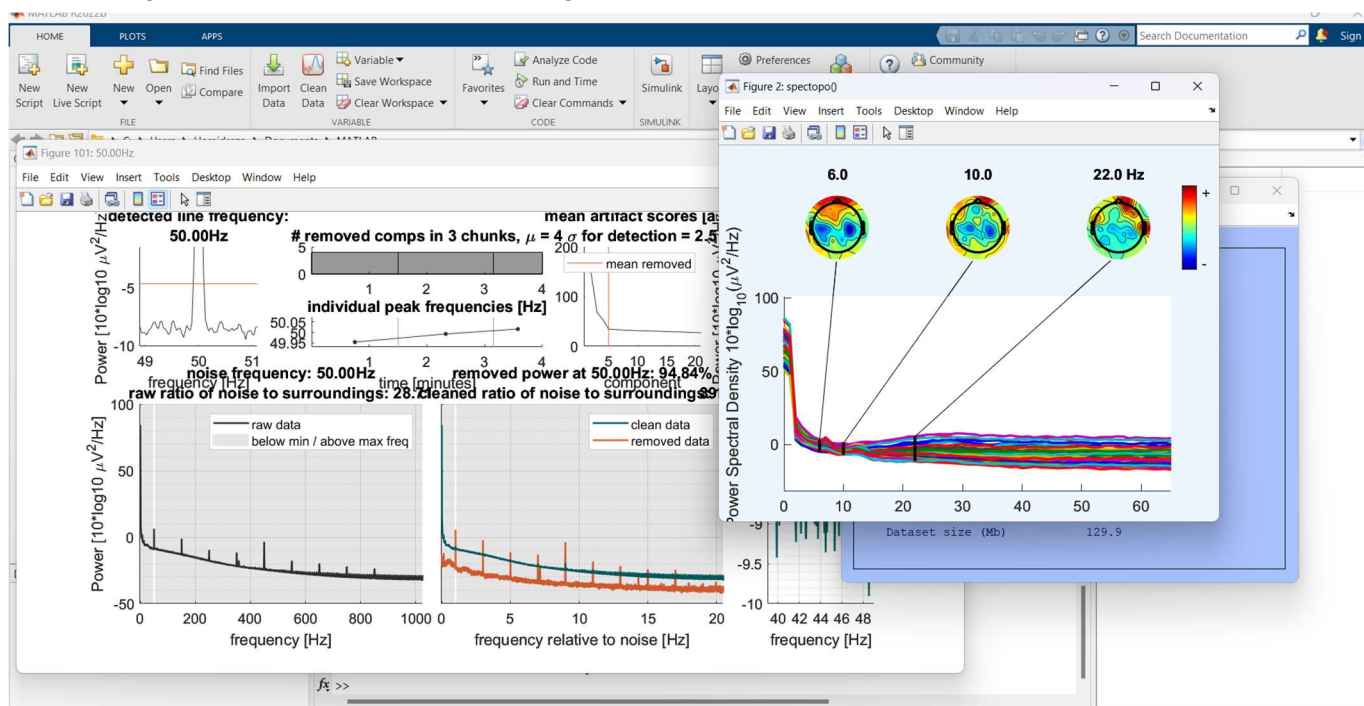


2# Channel placement





3# Checking for a bad channels and eliminating line noise.



pop_reref - average reference or re-reference data

Current data reference state is: unknown

☒ Compute average reference

☐ Huber average ref. with threshold: 25 uV

☐ Re-reference data to channel(s):

☐ Interpolate removed channel(s)

☐ Retain ref. channel(s) in data (will be flat for single-channel ref.)

Exclude channel indices (EMG, EOG):

Add old ref. channel back to the data:

Help Cancel Ok

Output data will be 2457 frames by 198 smoothed trials.
Outliers: 2.50 to 199.50
Not all sortvar values within time vector limits:
outliers will be shown at nearest limit.
Overplotting sorted sortvar on data.
Done.

Computing spectra (window length 2048; fft length: 2048; overlap 0):

Click on each trace for channel/component index
Note: Type "set(gcf, 'renderer', 'painter');" before saving the figure in postscript (eps) or jpg format

EEGLAB v2026.0.0

#2: project03_channel_zap

Filename: none
Channels per frame: 64
Frames per epoch: 491537
Epochs: 1
Events: none
Sampling rate (Hz): 2048
Epoch start (sec): 0.000
Epoch end (sec): 240.008
Reference: unknown
Channel locations: Yes
ICA weights: No
Dataset size (Mb): 129.9

4# Reference determination and High-low pass filter

What do you want to do with the new dataset?

Name it: project03_channel_zap_reref_highpas_lowpas Edit description

☐ Save it as file: Browse

Some changes have not been saved. What do you want to do with the old dataset?

☐ Overwrite it in memory (set=yes; unset=create a new dataset)

☐ Save it as file: Browse

Help Cancel Ok

length 2048; fft length: 2048; overlap 0):

nnel/component index

rer', 'painter');" before saving the figure i

et 3

ng 13519 point highpass filtering.

pop_eegfiltnew() - transition band width: 0.499955614735908 Hz

pop_eegfiltnew() - passband edge(s): 0.5 Hz

pop_eegfiltnew() - cutoff frequency(ies) (-6 dB): 0.250022192632046 Hz

pop_eegfiltnew() - filtering the data (zero-phase, non-causal)

firfilt(): |=====| 100%, ETE 00:00

Creating a new ALLEEG dataset 4

Done

pop_eegfiltnew() - performing 603 point lowpass filtering.

pop_eegfiltnew() - transition band width: 11.2265780730897 Hz

pop_eegfiltnew() - passband edge(s): 45 Hz

pop_eegfiltnew() - cutoff frequency(ies) (-6 dB): 50.6132890365448 Hz

pop_eegfiltnew() - filtering the data (zero-phase, non-causal)

firfilt(): |=====| 100%, ETE 00:00

f>

Phase

Phase (degrees) $\times 10^5$

Frequency (Hz)

Magnitude

Magnitude (dB)

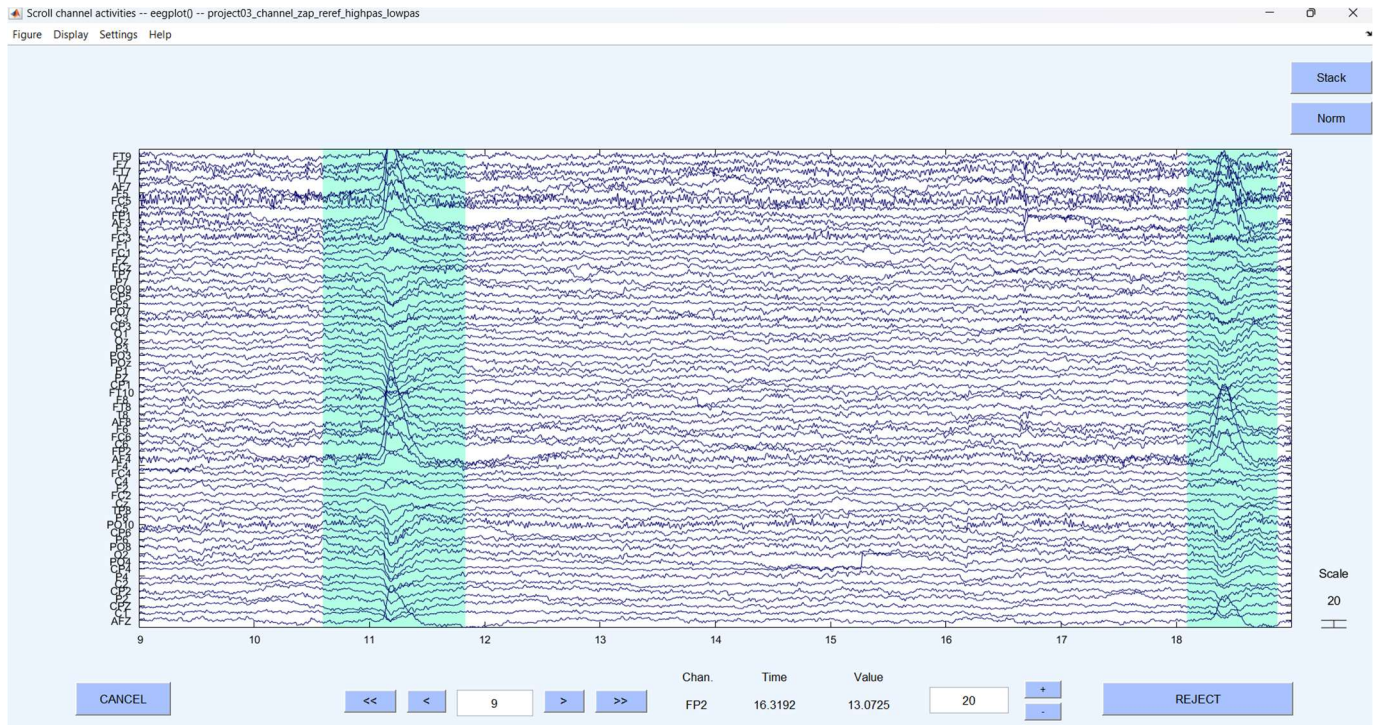
Frequency (Hz)

Phase

Phase (degrees)

Frequency (Hz)

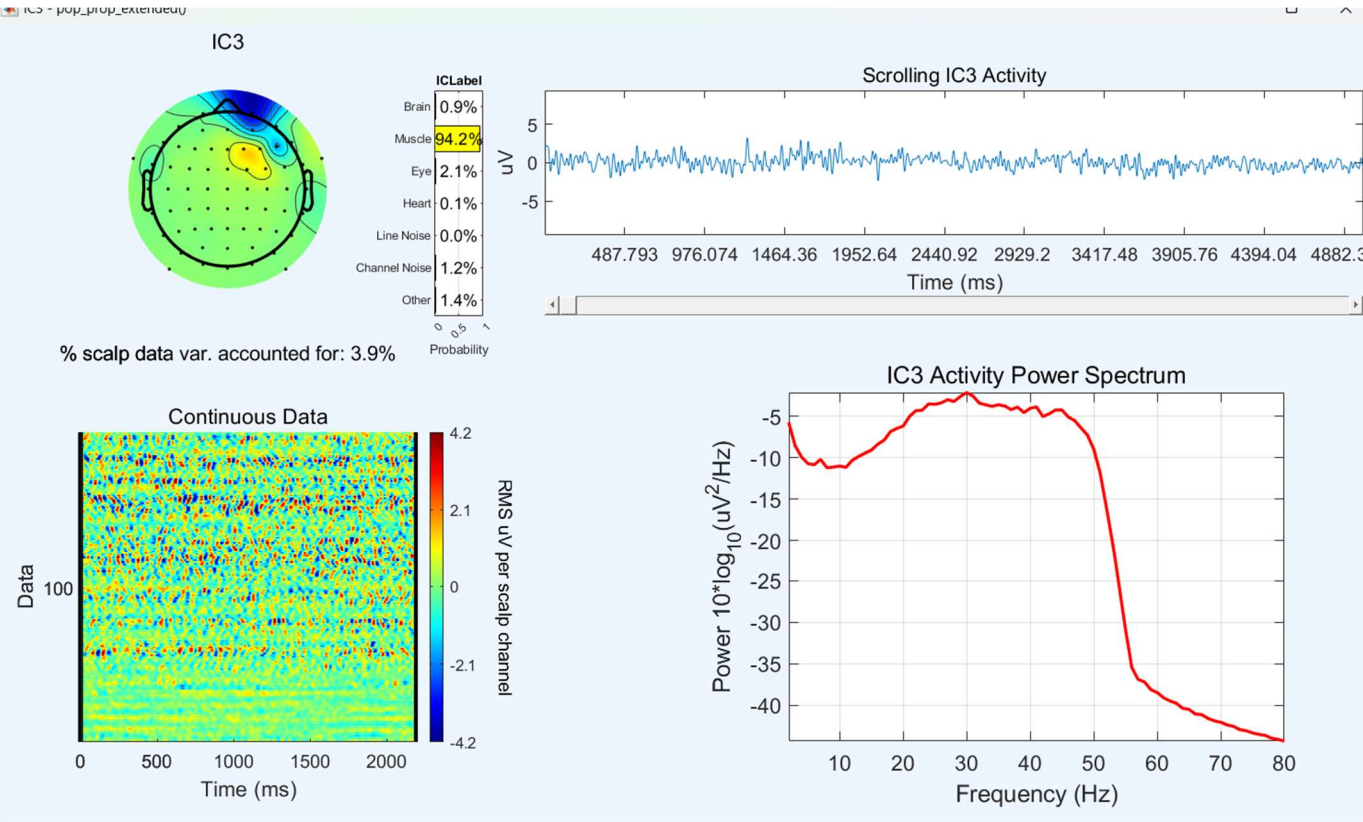
Reference: average
Channel locations: Yes
ICA weights: No
Dataset size (Mb): 129.9



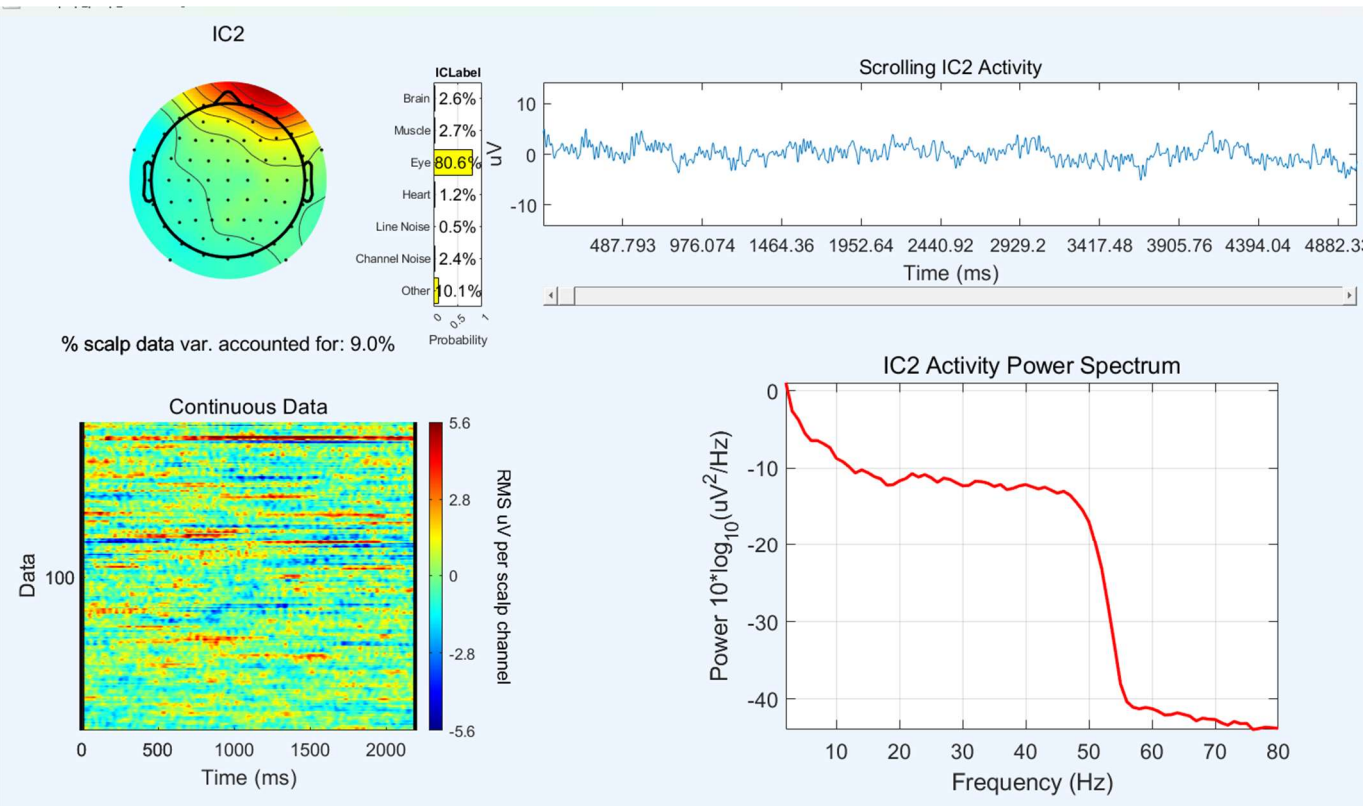
5# Eyes Rejecting Before the ICA Stage

PART 2 | ICA

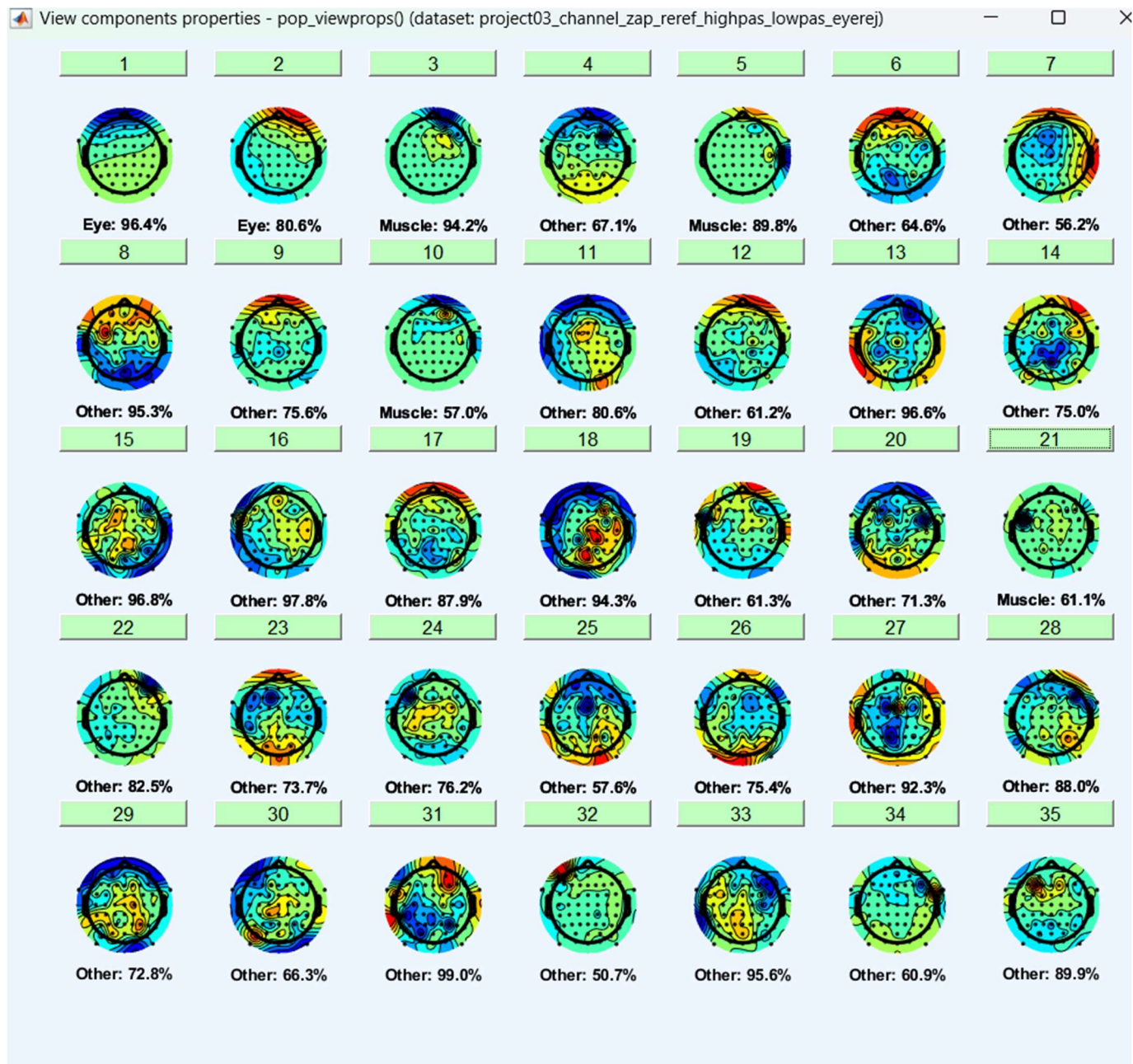
2. Then run ICA on the data and reject the noisy components.
3. Take a screenshot of the noisy components and explain why you considered them noisy.

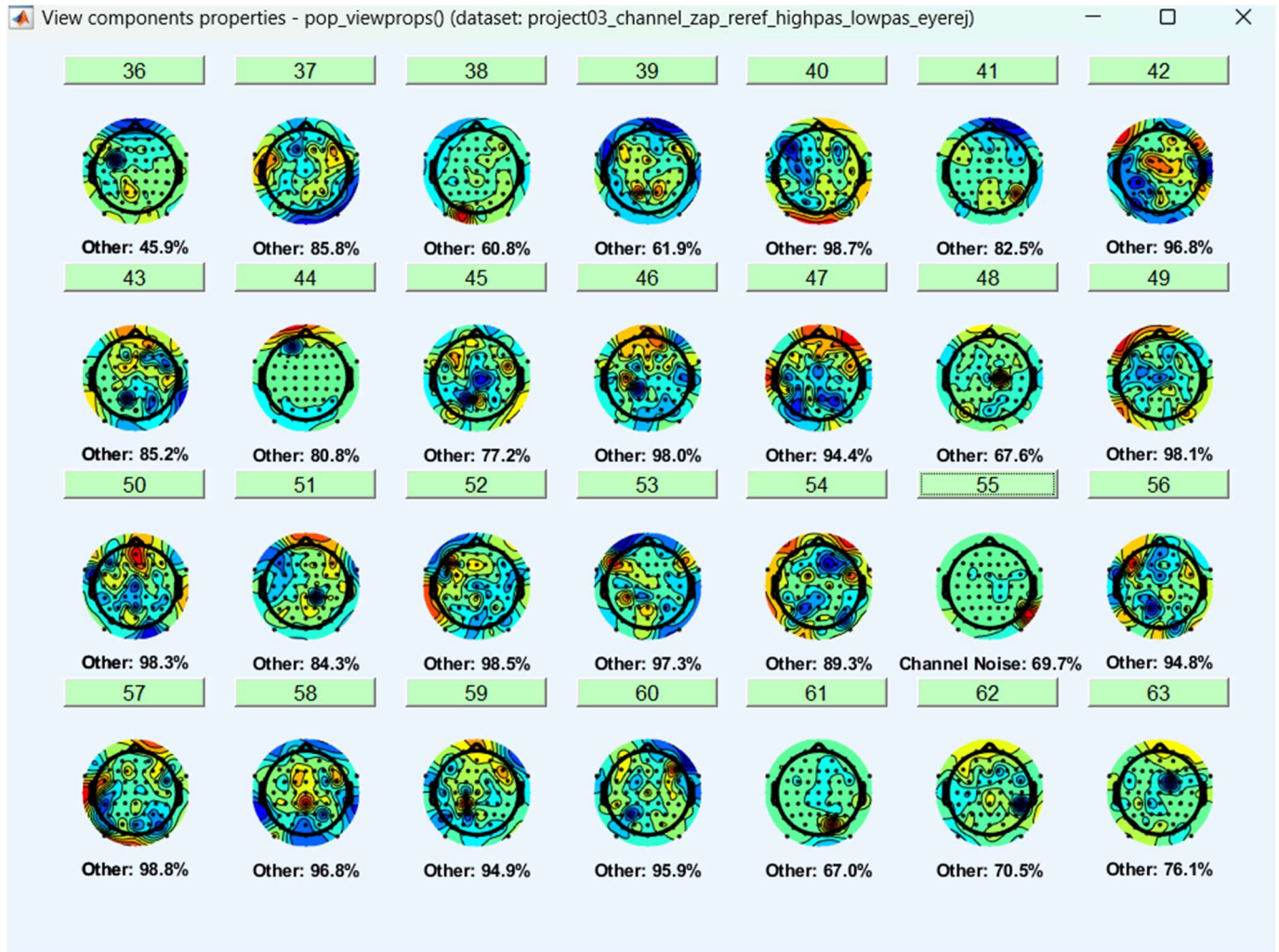


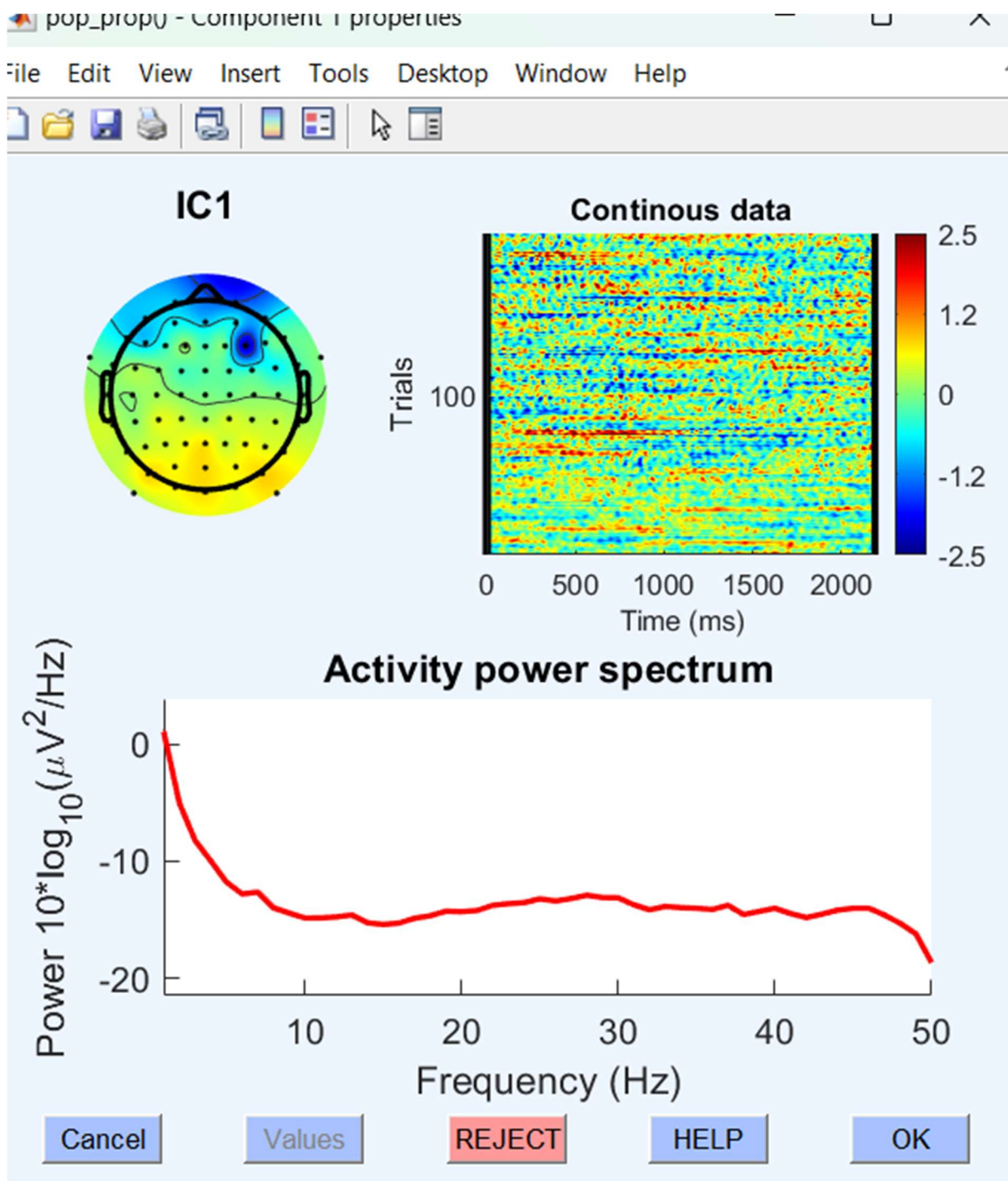
Muscle Component

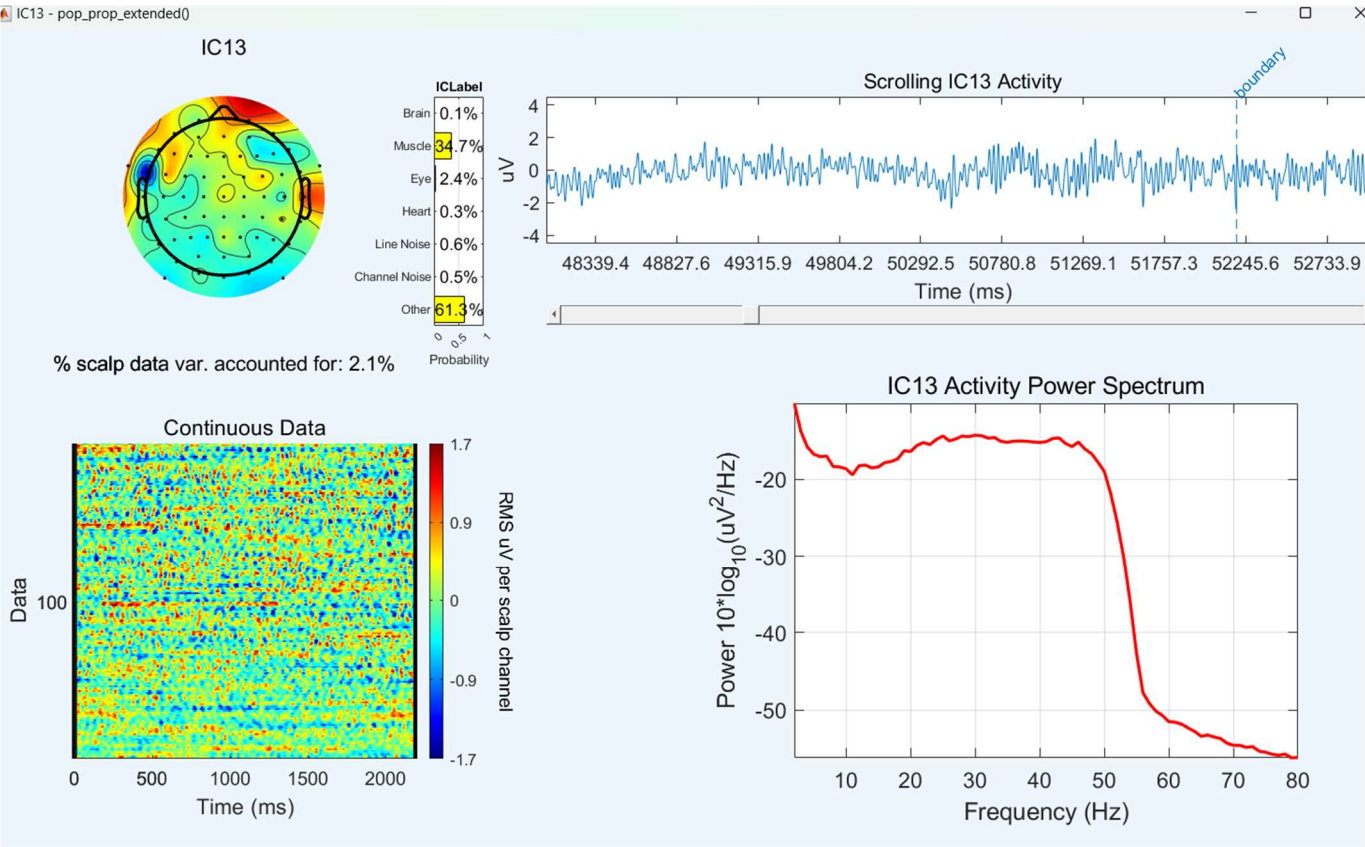


Eye Component

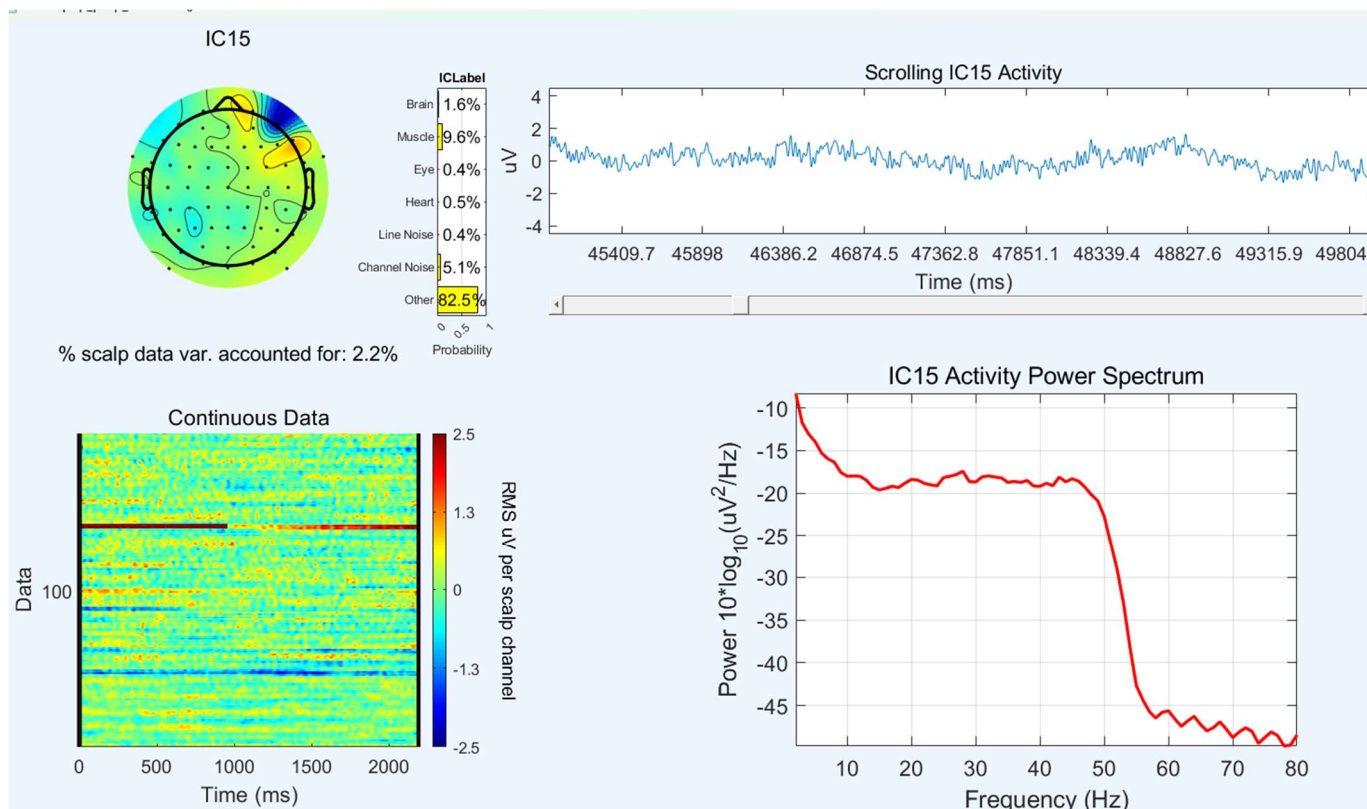




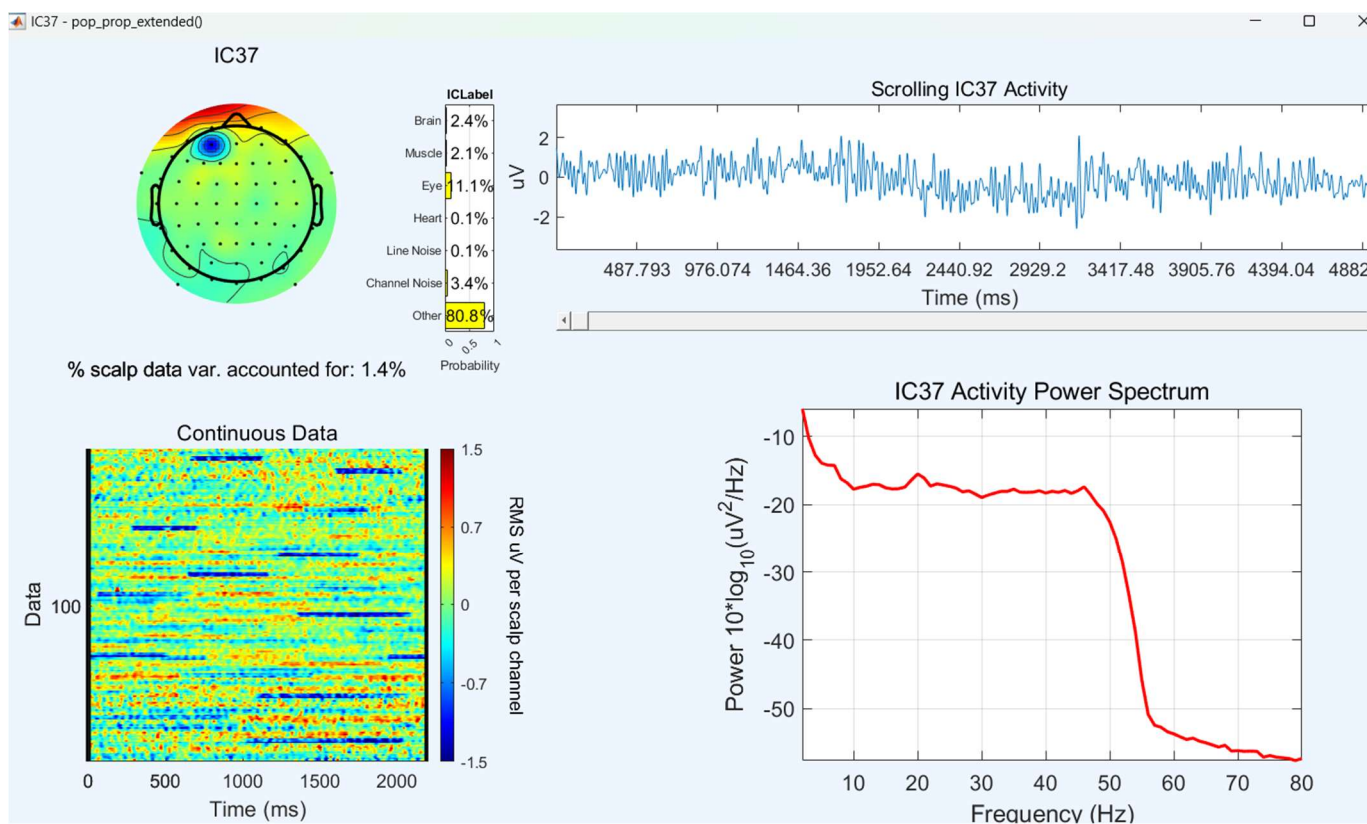




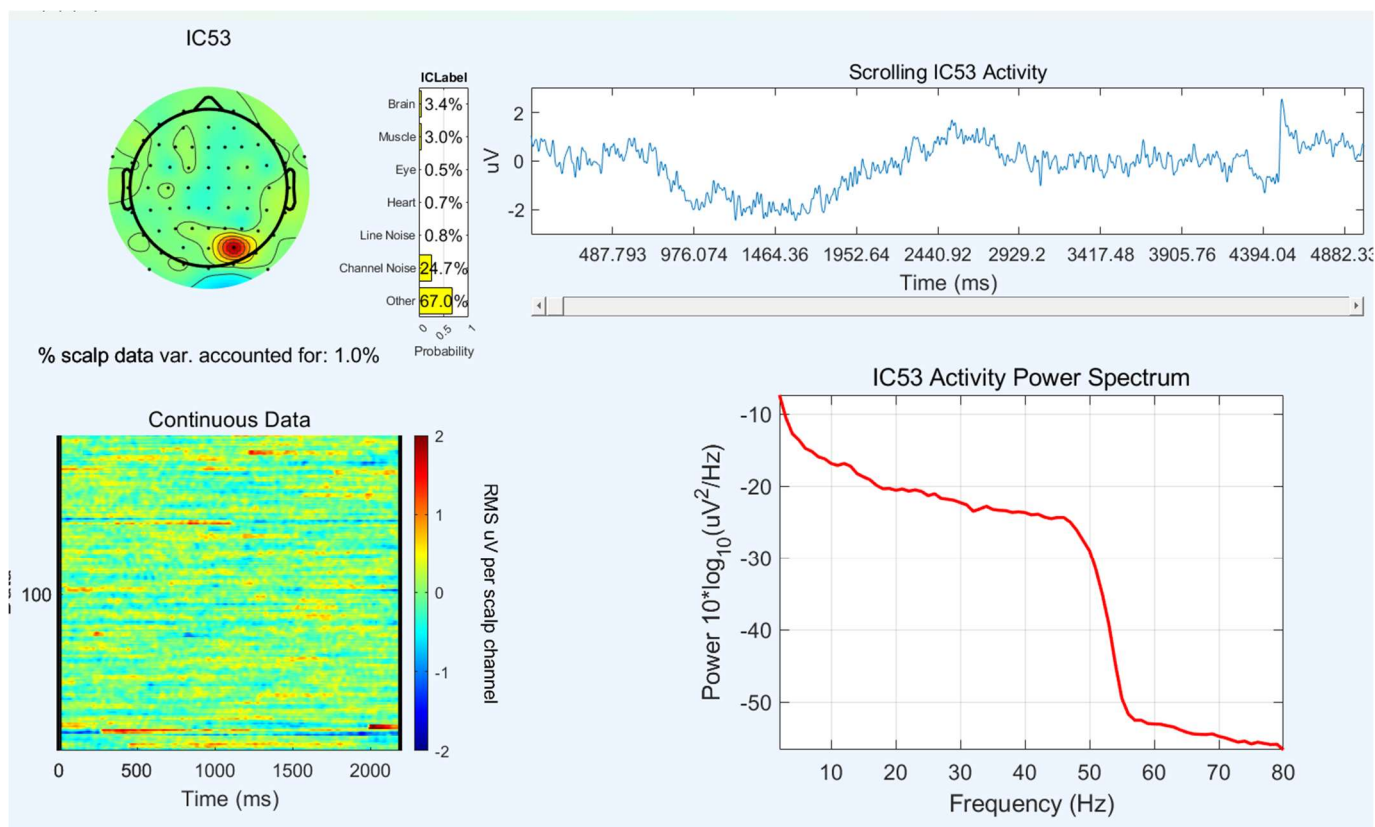
Muscle Component



Channel Noise



Muscle Component



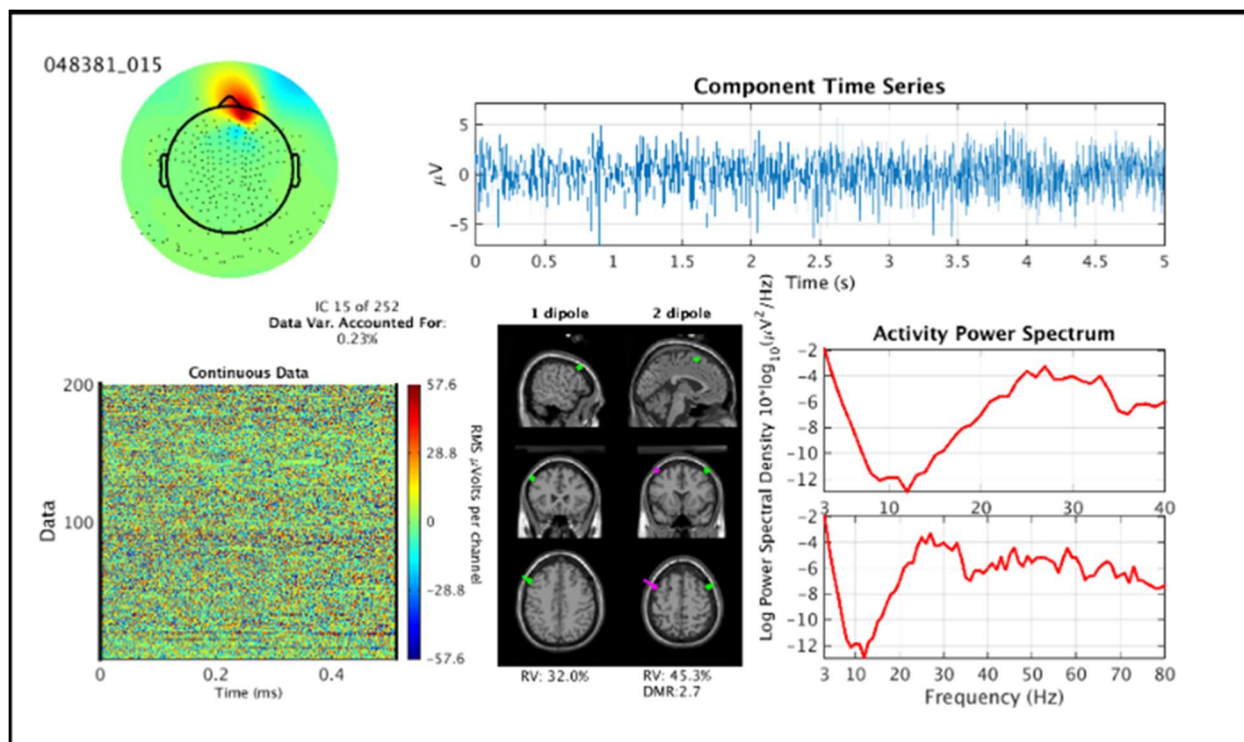
Channel Noise

PART 3 |

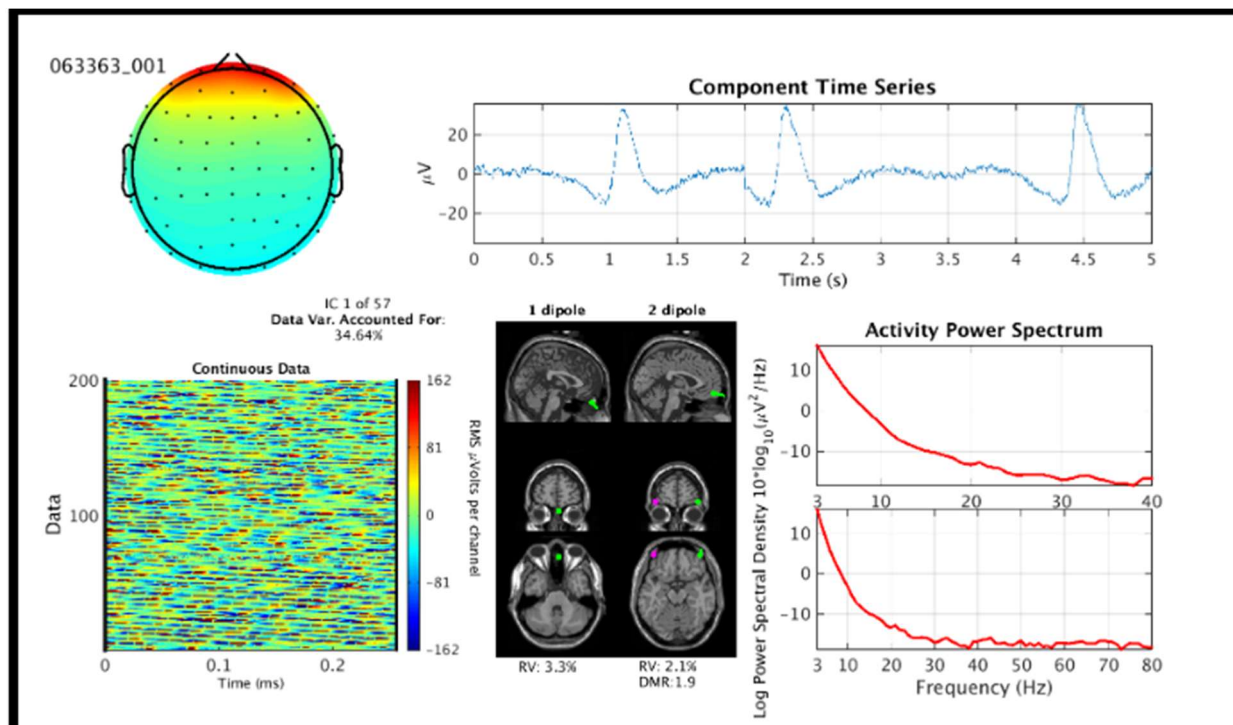
- Interpolate removed channels (if exist)
- Save the new dataset as project03_preprocessed_ICA.set.

PART 4 |

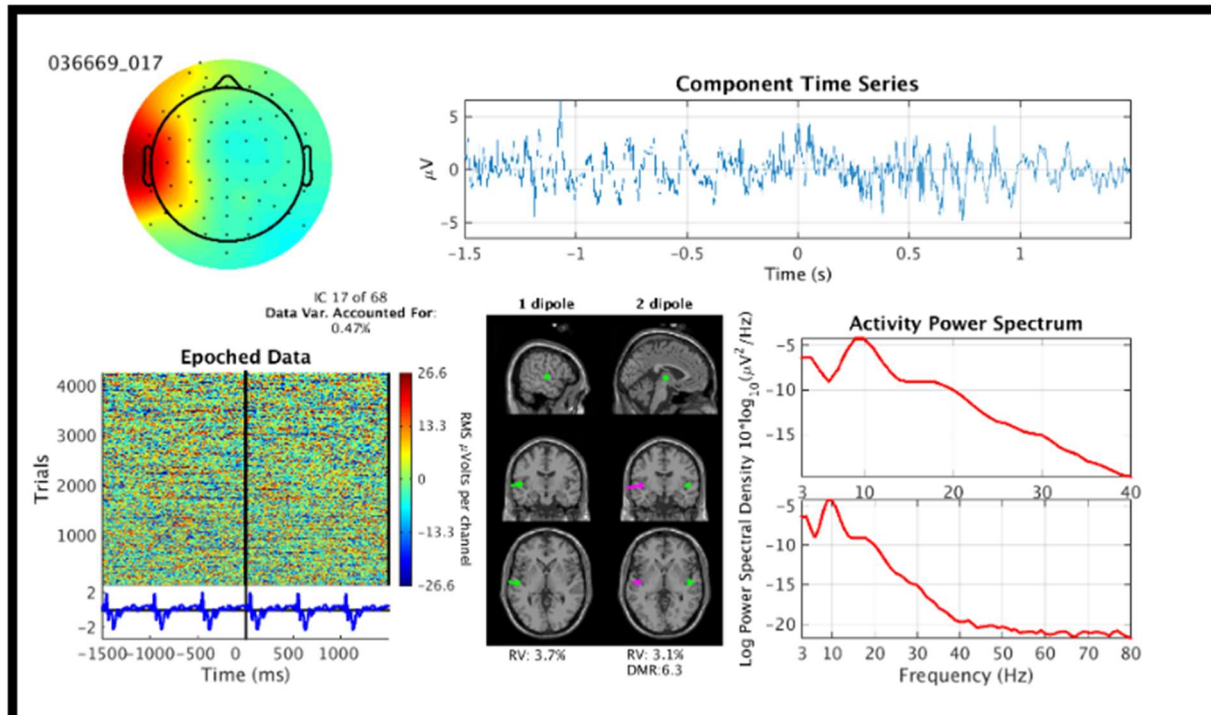
- Review the components below and explain what you think each one represents, along with the reasoning behind your interpretation.



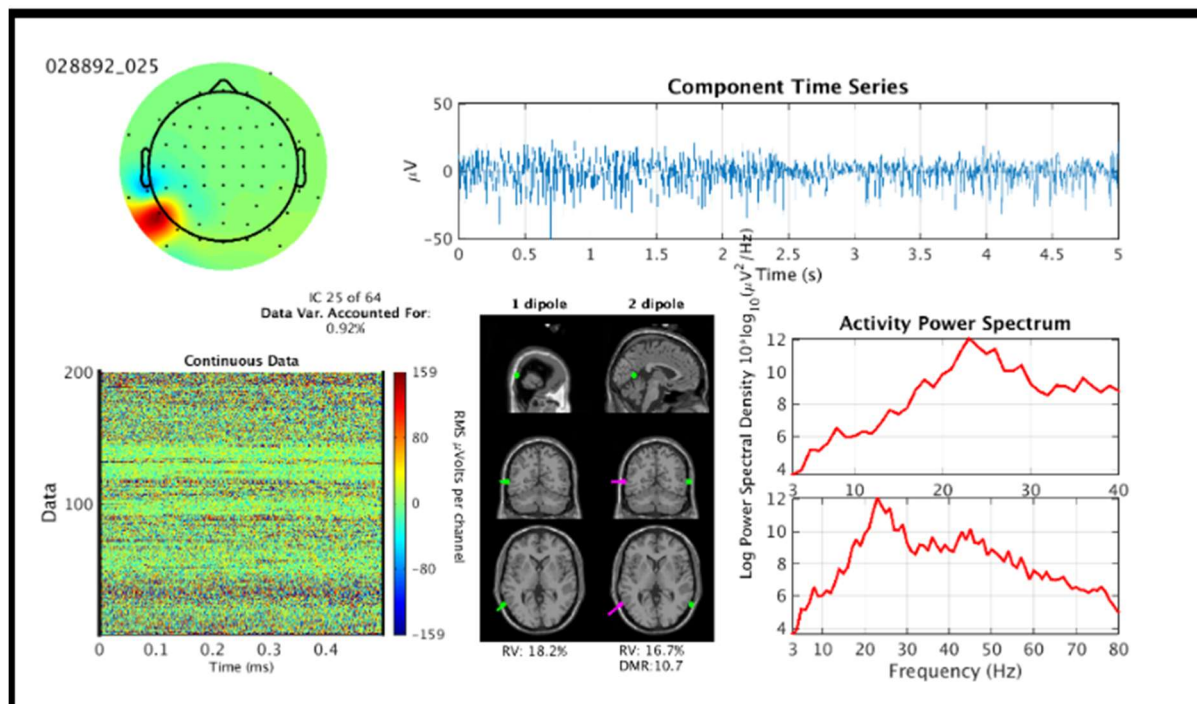
I think this is a muscle artifact because we can see a very strong increase in the amplitude of the component time series. There is also a prominent vertical pattern around the eye/eyelid region, and in the first plot, we can see very small, scattered points. In addition, the power spectrum shows an increasing trend toward higher frequencies, which is consistent with a muscle artifact.



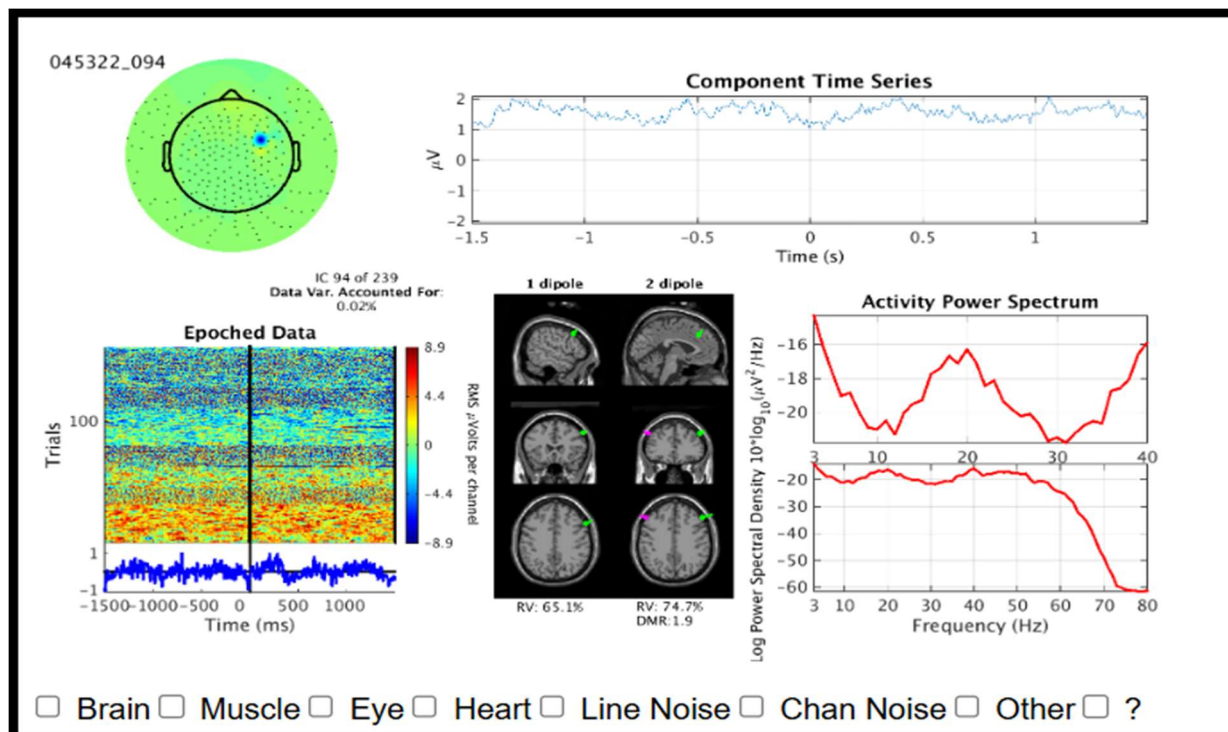
I think this is an ocular artifact, most likely an eye-blink artifact. We can see a lot of back-and-forth fluctuations in the component time series, and the activity is mainly concentrated in the frontal region.



In my opinion, this appears to be a neural component, and all the evidence supports this interpretation. In particular, we can see a clear peak around 10 Hz in the power spectrum, along with a gradual decrease toward lower frequencies. The component time series also appears to be relatively normal, without any obvious artifact-related patterns.



The high-frequency activity in the power spectrum, the fine, tooth-like pattern in the raster plot, the concentration of activity in one area, and the strong compression in the component time series all suggest that this is a muscle artifact.



It appears that we are dealing with some type of channel noise, mainly because the activity is highly concentrated in one location. The different frequencies seem to cancel each other out in the component time series, while the power spectrum shows a relatively flat, straight-line pattern. Overall, these features suggest that this is likely channel noise.